

# CHOOSECOLS + VSTACK in Excel

## 1. What is CHOOSECOLS + VSTACK?

CHOOSECOLS and VSTACK are two functions used together. CHOOSECOLS picks the columns you want from a table, in the order you want them, and VSTACK stacks tables one below the other. Together they solve the problem of combining tables that have extra columns or a different column order. A Data Analyst uses them to merge data from several sources into one clean table.

## 2. Syntax

`=VSTACK(CHOOSECOLS(array1, col_num1, [col_num2], ...), CHOOSECOLS(array2, col_num1, [col_num2], ...))`

| Function   | Argument      | Required? | What it accepts                                                                    |
|------------|---------------|-----------|------------------------------------------------------------------------------------|
| CHOOSECOLS | array         | Required  | The range to take columns from (for example A2:C3)                                 |
| CHOOSECOLS | col_num1      | Required  | The number of the first column to return. A negative number counts from the right. |
| CHOOSECOLS | col_num2, ... | Optional  | More column numbers, in the order you want them                                    |
| VSTACK     | array1        | Required  | The first range, or CHOOSECOLS result, to stack                                    |
| VSTACK     | array2, ...   | Optional  | More ranges or results, placed below the first                                     |

Excel works from the inside out: CHOOSECOLS selects the columns first, then VSTACK stacks the results.

## 3. Simple Example

Stack two sales tables, keeping only the ID and Product columns:

|   | A    | B          | C     | D | E    | F          |
|---|------|------------|-------|---|------|------------|
| 1 | ID   | Product    | Sales |   | ID   | Product    |
| 2 | P001 | Laptop     | 1200  |   | P001 | Laptop     |
| 3 | P002 | Mouse      | 150   |   | P002 | Mouse      |
| 4 |      |            |       |   | P004 | Monitor    |
| 5 | ID   | Product    | Sales |   | P005 | Headphones |
| 6 | P004 | Monitor    | 800   |   |      |            |
| 7 | P005 | Headphones | 250   |   |      |            |

Formula in E2 only (headings in E1:F1 are typed):

`=VSTACK(CHOOSECOLS(A2:C3, 1, 2), CHOOSECOLS(A6:C7, 1, 2))`

The first table is in A1:C3 and the second in A5:C7. CHOOSECOLS keeps columns 1 and 2 of each and drops Sales. VSTACK puts the second result below the first. **Result: 4 rows and 2 columns**

## 4. Another Example

Stack online and store orders. The store table has Amount and Customer in a different order:

|   | A     | B        | C        | D | E     | F        | G      |
|---|-------|----------|----------|---|-------|----------|--------|
| 1 | Order | Customer | Amount   |   | Order | Customer | Amount |
| 2 | O101  | Ayesha   | 250      |   | O101  | Ayesha   | 250    |
| 3 | O102  | Rahim    | 120      |   | O102  | Rahim    | 120    |
| 4 |       |          |          |   | O201  | Nabila   | 80     |
| 5 | Order | Amount   | Customer |   | O202  | Tanvir   | 300    |
| 6 | O201  | 80       | Nabila   |   |       |          |        |
| 7 | O202  | 300      | Tanvir   |   |       |          |        |

Formula in E2 only (headings in E1:G1 are typed): `=VSTACK(A2:C3, CHOOSECOLS(A6:C7, 1, 3, 2))`

The online orders (A2:C3) are already in the right order. For the store orders, CHOOSECOLS returns columns 1, 3, 2, which puts Customer before Amount. Now both parts match, and VSTACK stacks them.

## 5. When to Use It

- Combine monthly or regional tables, keeping only the columns you need
- Merge tables from different sources where the column order is different
- Build a master list of customers, orders, or products from several tables
- Prepare a compact, consistent dataset for a report or dashboard

## 6. Important Things to Remember

- Read the formula from the inside out: CHOOSECOLS selects the columns, then VSTACK stacks the results.
- Every part should have the same columns in the same order. If one has fewer columns, the missing cells show #N/A.
- Column numbers count inside each array. A negative number counts from the right. A 0 or a number that is too large gives #VALUE!.
- Include the heading row from the first table only, or type the headings yourself.
- The result spills into neighboring cells. They must be empty, or Excel shows #SPILL!.
- Both functions are available in Microsoft 365 and Excel 2024, but not in Excel 2021 or earlier.

## 7. Common Mistakes

**Mistake:** Selecting a different number of columns from each table, so #N/A appears. **Fix:** Choose the same number of columns from every table.

**Mistake:** Stacking tables with different column orders, so values land under the wrong headings. **Fix:** Use CHOOSECOLS to put the columns in the same order first.

**Mistake:** Including the heading row of the second table, so the headings repeat in the middle. **Fix:** Start each later range at its first data row.

## 8. Data Analyst Tip

Wrap the formula in another function. For example, `=SORT(VSTACK(A2:C3, CHOOSECOLS(A6:C7, 1, 3, 2)), 3, -1)` stacks the orders and sorts them by Amount, highest first.